Occlusion-Consistent Bidirectional RGB-D Geometric Fusion by an Iterated Kalman Smoother

A mathematical formulation and conditional convergence analysis

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Abstract

We study an offline sequence of calibrated multiview RGB-D observations of an articulated or piecewise-rigid scene. The objective is to construct *one* canonical point/surfel map and a sequence of shared rigid-body poses whose reprojections explain observed color, depth, free space, and occlusions. Initially visible points constrain a forward rigid-motion estimation; later-revealed points are transported backward and probabilistically tested against earlier depth images. The essential decomposition alternates (A) a continuous, analytic-Jacobian, error-state iterated extended Kalman smoother (IEKS) with fixed geometric hypotheses and (B) a deterministic probabilistic update of geometry existence, associations, and part labels, keeping the rigid trajectory fixed. We write a normalized ray observation model, show how both stages optimize a *single* posterior variational objective, and prove monotone improvement and convergence of objective values. Under additional smoothness, boundedness, gauge-fixing, and sufficiently improving block updates, accumulation points are stationary for the represented variational problem. We also prove a one-view ambiguity and explain why neither global-optimum recovery nor complete reconstruction of never-observed geometry follows from these results. No hand-selected geometric residual threshold enters the mathematical decisions.

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1 Problem statement and scope

Data. We have synchronized or timestamped views $c \in \{1, \dots, C_k\}$ at times $t_0 < \cdots < t_K$. Intrinsics K_c and world-to-camera poses $G_c = (R_{cw,c}, t_{cw,c})$ are known and fixed. Each valid pixel ray $r = (k, c, u)$ provides the measured RGB-D vector $y_r = (I_{k,c}(u), D_{k,c}(u))$; invalid-depth rays are retained as a separate measurement type. Its world-frame depth backprojection is

$$y_r^{3D} = G_c^{-1} \begin{bmatrix} D_{k,c}(u) K_c^{-1} \tilde{u} \\ 1 \end{bmatrix}, \quad \tilde{u} = (u_x, u_y, 1)^\top, \tag{1}$$

where homogeneous notation is used only in this equation. Camera RGB and depth are *joint readings of one ray*, not independent images to be counted twice.

Aim. Recover one canonical map $\mathcal{M}$ and a common history $\mathcal{T} = \{T_{b,k} \in \mathrm{SE}(3)\}$, such that the complete RGB-D images are likely under simple point projection, foreground ordering, and sensor noise. Crucially, a point that appears late may have been physically present but occluded at the beginning.

Deliberate exclusions. There is no 3D Gaussian Splatting renderer, no free per-frame nonrigid displacement of every point, no hard nearest-neighbor cutoff for deciding that a point is "new," and no assertion that all physical surfaces are identifiable.

2 Unified latent state and candidate geometry

2.1 Fixed finite hypothesis universe

Let $i \in \{1, \dots, N_{\max}\}$ index a finite *candidate catalogue*: any valid observed depth pixel may contribute a potential surface element, together with other candidates if available. A catalogue entry does *not* mean an accepted point; it is only a possible explanation. This construction permits newly revealed points without changing the mathematical dimension during optimization. Candidates may be initialized by backprojecting their source observations, but the original observation is inserted in the likelihood *once only*.

Each candidate has canonical location $p_i \in \mathbb{R}^3$, appearance $a_i \in \mathbb{R}^3$, an optional local tangent n_i, and a known or estimated projected support derived from RGB-D pixel footprint and local surface geometry. Let $e_i \in \{0, 1\}$ indicate its existence; $b_i \in \{0, \dots, B\}$ denotes its rigid component, with $b = 0$ fixed in the world. Both variables are unknown. The finite B is a modeled maximum, not an assertion about the true number of parts; a prior can favor unused components.

Pose $T_{b,k} = (R_{b,k}, t_{b,k})$ maps canonical component b coordinates into world coordinates at frame k:

$$x_{i,k} = T_{b_i,k} p_i = R_{b_i,k} p_i + t_{b_i,k}. \tag{2}$$

Fix one reference pose per component (e.g. $T_{b,0} = I$) to remove the canonical-coordinate gauge; this is a coordinate convention even if that component is not visible at time zero. Unused components may remain unidentifiable.

2.2 Unknowns and two optimization blocks

Write $s = \{T_{b,k}, \nu_{b,k}\}_{b,k}$ for rigid poses and optional twists; $m = \{p_i, a_i, n_i\}_i$ for continuous canonical map parameters; and $z = (\{e_i, b_i\}_i, \{h_r\}_r)$ for latent existence, part labels, and per-ray generating surface $h_r \in \{0, 1, \dots, N_{\max}\}$. The *motion* block s is estimated with m and the probabilistic responsibilities held fixed. The *geometry/association* block updates its posterior and m while s is fixed. Both blocks are judged by one common objective.

3 Direct point reprojection and analytic motion Jacobians

For candidate i at time k in camera c define

$$m_{i,r} = R_{cw,c}(R_{b_i,k}p_i + t_{b_i,k}) + t_{cw,c}, \quad u_{i,r} = \pi_c(m_{i,r}), \quad d_{i,r} = e_3^\top m_{i,r}, \tag{3}$$

with the usual pinhole projection $\pi(X,Y,Z) = (f_xX/Z + c_x, f_yY/Z + c_y)$. For a valid predicted-visible association, the *sampled-image* residual is

$$r_{i,r}(s,m) = \begin{bmatrix} I_{k,c}(u_{i,r}) - a_i \\ D_{k,c}(u_{i,r}) - d_{i,r} \end{bmatrix} \in \mathbb{R}^4. \tag{4}$$

This is a useful *conditional registration residual*; it is not by itself a complete generative likelihood for an entire image (Section 4). Values and gradients are interpolated only where valid measured samples support interpolation. A surfel is not required to coincide with a sampled pixel center.

For a left Lie-algebra world-frame perturbation $T_{b,k}^{+} = \mathrm{Exp}(\widehat{\delta\xi})T_{b,k}$, $\delta\xi = (\delta v, \delta\omega)$, the Jacobian of world position is

$$\delta x_{i,k} = J_x\delta\xi, \qquad J_x = [\, I_3 \quad -[x_{i,k}]_\times \,]. \tag{5}$$

Let $J_\pi = \partial\pi/\partial m$. Then the exact local derivatives of (4) within an interpolation cell are

$$H_{i,r}^{I} = \nabla I(u_{i,r})J_\pi R_{cw,c}J_x, \tag{6}$$

$$H_{i,r}^{D} = (\nabla D(u_{i,r})J_\pi - e_3^\top)R_{cw,c}J_x. \tag{7}$$

The canonical-point derivative instead uses $\partial x/\partial p = R_{b,k}$. Motion Jacobians have only six columns per rigid body pose, not twelve or more columns per scene point. With source time s_i and inverse transport, derivatives involve both $T_{b,k}$ and T_{b,s_i}: the source point is not an immovable estimate once joint smoothing begins.

3.1 Forward and backward transport are the same map

If measurement $y_{i,s}^{3D}$ at time s initializes candidate i under part hypothesis b,

$$p_i^{(b)} = T_{b,s}^{-1}y_{i,s}^{3D}, \qquad \widehat{x}_{i,k}^{(b)} = T_{b,k}T_{b,s}^{-1}y_{i,s}^{3D}. \tag{8}$$

Equation (8) works for $k > s$ and $k < s$, and for nonconsecutive timestamps. A point born at the final frame is merely a point with source index $s_i = K$. Its uncertainty must include uncertainty in T_{b,s_i}; preferably its canonical position remains an optimized shared variable rather than a permanently frozen inverse warp.

4 A normalized RGB-D ray model with occlusion

4.1 Geometric coverage and ordered visibility

A bare point has zero image area. Give each candidate a small oriented surface support S_i (a *surfel*), with its extent tied to an RGB-D pixel footprint/local depth sampling model; this is a finite geometric element, not a 3DGS Gaussian. Let $c_{r,i}(s,m) \in [0,1]$ denote the probability that surface i covers ray/pixel r, computed by projected footprint overlap convolved with the calibrated pixel response. Its dependence on geometry can be differentiated wherever smooth. Set $c_{r,i} = 0$ for projections outside the camera or behind its valid depth range. Condition on existence and component assignments e, b.

Order the projected candidates on this ray by increasing positive camera depth. In a region where this ordering is fixed, the generative probability that candidate i is the first covered element is

$$v_{r,i} = e_i c_{r,i} \prod_{j:\, d_{j,r} < d_{i,r}} (1 - e_j c_{r,j}), \qquad v_{r,\emptyset} = \prod_j (1 - e_j c_{r,j}). \tag{9}$$

For fixed binary existences these are normalized competing-first-hit probabilities:

$$v_{r,\emptyset} + \sum_i v_{r,i} = 1. \tag{10}$$

A later point is invisible if a closer foreground element almost certainly covers the ray. Occlusion is therefore asymmetric and is explained by measured line-of-sight geometry rather than by a fitted distance cutoff. On depth-order ties the map is piecewise smooth; the convergence theorem will explicitly exclude nonregular ties at its limit points (or require a smooth probabilistic ordering model).

4.2 The single-ray sensor likelihood

For a valid measured RGB-D ray define $f_{r,i}(y_r \mid s, m)$ as the normalized sensor density centered at $(a_i, d_{i,r})$. For example, for *one* four-dimensional joint sensor noise model,

$$f_{r,i}(y_r) = \mathcal{N}_4\left(y_r; \begin{bmatrix} a_i \\ d_{i,r} \end{bmatrix}, R_{r,i}\right), \quad R_{r,i} \succ 0. \tag{11}$$

Use a calibrated joint covariance or a measured photometric correction if needed. For an empty/model-unexplained ray use a proper density $f_{r,\emptyset}(y_r)$ (such as a broad calibrated background/novelty distribution, *not* an arbitrary uniform score). Then

$$\boxed{L_r(y_r \mid s, m, e, b) = v_{r,\emptyset} f_{r,\emptyset}(y_r) + \sum_i v_{r,i} f_{r,i}(y_r).} \tag{12}$$

Missing-depth observations require their own explicit probability mass (including sensor failure and no-return mechanisms), not a fictitious zero depth. One may augment each $f_{r,i}$ with a calibrated contamination/no-return component, so valid gross outliers are not automatically interpreted as birth events. Data noise parameters and priors are *part of the statistical model*; rejecting a point is never based on a post-hoc depth/RGB threshold.

Remark 4.1 (Two complementary expressions)**.** Equation (4) is a dense registration residual obtained by sampling the *observed* images at projected candidate locations and is convenient for fast analytic IEKS updates. Equation (12) is a normalized *generative ray likelihood* of the full images. To prove convergence for the full posterior, stage A must differentiate and line-search the objective induced by (12), including geometric coverage/visibility terms, or accept a proposed IEKS step only after checking that this actual common objective increases. Decreasing an arbitrary sum of the conditional residuals alone does *not* establish the full theorem.

4.3 Why an occluded candidate does not produce a photometric error

Consider a target depth d and a candidate behind a closer well-supported element at $d_f < d_i$. As $c_{r,f} \to 1$, equation (9) gives $v_{r,i} \to 0$: color a_i and depth d_i do not influence L_r. Conversely, a candidate in front of the observed surface can become the predicted first hit; its mismatch changes the likelihood. This is a *ray likelihood comparison*, not a symmetric point-to-cloud nearest-neighbor penalty.

5 Temporal process model, IEKF and smoothing

5.1 Rigid-body dynamics as a statistical prior

Let the dynamic block of body b include $T_{b,k} \in \mathrm{SE}(3)$ and a twist $\nu_{b,k} \in \mathbb{R}^6$. One possible neutral process is

$$T_{b,k+1} = \mathrm{Exp}(\Delta t_k \widehat{\nu}_{b,k}) T_{b,k} \,\mathrm{Exp}(\widehat{w}_{b,k}^T), \quad \nu_{b,k+1} = \nu_{b,k} + w_{b,k}^{\nu}, \tag{13}$$

where joint process noise has a specified positive-definite covariance $Q_{b,k}$ in local coordinates; the exact order of the right pose perturbation is a convention to be reflected in the Jacobian. An articulated rigid-motion prior may replace free body twist if a constant joint model is already justified. No assumption of constant velocity between arbitrarily distant observations is required beyond the chosen Q and actual time separation.

The negative log process prior is $E_{\mathrm{dyn}}(s) = \frac{1}{2}\sum_{b,k} \left\|\mathrm{Log}(F_{b,k}(s)^{-1}T_{b,k+1})\right\|^2_{Q_{b,k}^{-1}}$ with the analogous twist terms. A prior on the gauge-fixed initial state anchors scale and geometry.

5.2 Conditional motion update

For fixed m and a distribution $q(z)$ of geometry/association hypotheses, define the *motion objective*

$$F_A(s \mid m, q) = \mathbb{E}_q[\log p(\mathcal{Y}, z \mid s, m)] + \log p(s) + \mathrm{const}, \tag{14}$$

where (12) determines the complete-data measurement terms. Its local linearization yields gradient $g_s = \nabla_s(-F_A)$ and a symmetric positive-definite damped information approximation

$$B_s = \Lambda_{\mathrm{dyn}} + J_s^\top W J_s + \lambda I \succ 0, \quad B_s\Delta_s = -g_s. \tag{15}$$

Here W is based on measurement uncertainty and current latent responsibilities. Contributions of the derivatives of $\log v_{r,i}$ must be included in g_s; if a Gauss–Newton approximation omits some second derivatives, it is still valid as a search metric so long as the *full gradient* is used. A line search is performed on the *actual* objective F_A (or common ELBO below), not on a frozen-visibility surrogate. *This is an information-form IEKF/IEKS step, not generic point-position gradient descent.*

5.3 Back-and-forth without data double counting

Forward IEKF and Rauch–Tung–Striebel backward smoothing are two computational methods for the *same* fixed-likelihood trajectory estimation, not two independent likelihood factors. If $P_{k|k}$ is the filtered covariance, F_k the linearized transition and $P_{k+1|k}$ the one-step prediction covariance, the local Euclidean smoother uses

$$G_k = P_{k|k}F_k^\top P_{k+1|k}^{-1}, \qquad \hat{s}_{k|K} = \hat{s}_{k|k} + G_k(\hat{s}_{k+1|K} - \hat{s}_{k+1|k}). \tag{16}$$

On SE(3), pose differences and additions are replaced by the appropriate Lie logarithms and retractions. Repeated iterations *relinearize the original complete time-series objective*; they must not use the already updated posterior as a new independent prior while reintroducing the same RGB-D pixels.

6 Explicit nonlinear EKF state and output equations

This section states the filtering model in implementable state-space form. The original process (13) and ray likelihood (12) are its dynamics and observation laws, respectively. The distinctions between a conditional Gaussian reprojection and the full occlusion-mixture output are essential.

6.1 State, errors, and exact declared dynamics

For each moving component $b = 1, \ldots, B$, define the world-frame spatial twist $\nu_{b,k} = (v_{b,k}, \omega_{b,k}) \in \mathbb{R}^6$ and canonical-to-world pose $T_{b,k} \in \mathrm{SE}(3)$. The time-dependent filtering state is

$$s_k = ((T_{1,k}, \nu_{1,k}), \ldots, (T_{B,k}, \nu_{B,k})), \qquad \delta s_k \in \mathbb{R}^{12B}. \tag{17}$$

The canonical map $m = \{p_i, a_i, S_i\}$ is time-invariant; point existence and part assignment are separate latent scene variables. Static component $b = 0$ has $T_{0,k} = I$. In translation-first

coordinates $\xi = (\rho, \phi)$, write $\widehat{\xi} = \begin{bmatrix} [\phi]_\times & \rho \\ 0 & 0 \end{bmatrix}$. Use a left pose error and an additive world-frame twist error:

$$T_{b,k} = \mathrm{Exp}(\widehat{\delta\xi}_{b,k})\bar{T}_{b,k}, \quad \nu_{b,k} = \bar{\nu}_{b,k} + \delta\nu_{b,k}, \quad \delta s_{b,k} = \begin{bmatrix} \delta\xi_{b,k} \\ \delta\nu_{b,k} \end{bmatrix}. \tag{18}$$

The explicit nonlinear state equation, including the *right-applied* local pose process noise chosen in (13), is

$$\boxed{\begin{aligned} T_{b,k+1} &= \mathrm{Exp}(\Delta t_k \widehat{\nu}_{b,k}) T_{b,k} \,\mathrm{Exp}(\widehat{w}^T_{b,k}), \\ \nu_{b,k+1} &= \nu_{b,k} + w^\nu_{b,k}, \quad (w^T_{b,k}, w^\nu_{b,k}) \sim \mathcal{N}(0, Q^{\mathrm{local}}_{b,k}). \end{aligned}} \tag{19}$$

Set $\varphi = \Delta t_k \bar{\nu}_{b,k}$, $A = \mathrm{Exp}(\widehat{\varphi})$ and $\bar{T}' = A\bar{T}_{b,k}$. The mean prediction is $\bar{T}_{b,k+1|k} = \bar{T}'$, $\bar{\nu}_{b,k+1|k} = \bar{\nu}_{b,k|k}$. With the translation-first adjoint $\mathrm{Ad}_{(R,t)} = \begin{bmatrix} R & [t]_\times R \\ 0 & R \end{bmatrix}$ and the *left* group Jacobian $J_l(\varphi) = \int_0^1 \exp(\tau \,\mathrm{ad}_\varphi)\, d\tau$, the first-order transition is

$$\boxed{F_{b,k} = \begin{bmatrix} \mathrm{Ad}_A & \Delta t_k J_l(\varphi) \\ 0 & I_6 \end{bmatrix}, \quad G_{b,k} = \begin{bmatrix} \mathrm{Ad}_{\bar{T}'} & 0 \\ 0 & I_6 \end{bmatrix}.} \tag{20}$$

Indeed, $A\,\mathrm{Exp}(\widehat{\delta\xi})\bar{T} = \mathrm{Exp}(\widehat{\mathrm{Ad}_A\,\delta\xi})A\bar{T}$, while $\mathrm{Exp}(\widehat{\varphi + \Delta t \delta\nu}) = \mathrm{Exp}(\widehat{J_l(\varphi)\Delta t \delta\nu})A + O(\|\delta\nu\|^2)$; the right noise $\bar{T}'\,\mathrm{Exp}(\widehat{w}^T)$ becomes left error $\mathrm{Ad}_{\bar{T}'}\, w^T$. Stack these blocks, retaining any modeled cross-body covariance. Thus

$$\boxed{\hat{s}^-_{k+1} = f_k(\hat{s}^+_k, 0), \qquad P^-_{k+1} = F_k P^+_k F_k^\top + G_k Q^{\mathrm{local}}_k G_k^\top.} \tag{21}$$

This covariance equation is *not* equivalent to $FPF^\top + Q^{\mathrm{local}}$ when the process noise is right-applied as above.

6.2 Explicit RGB-D output equation and its analytic Jacobian

For candidate i assigned to component b_i, at time k and camera c, define

$$x_{i,k} = T_{b_i,k} p_i, \quad m_{i,c,k} = R_{cw,c} x_{i,k} + t_{cw,c}, \quad u_{i,c,k} = \pi_c(m_{i,c,k}), \quad d_{i,c,k} = e_3^\top m_{i,c,k}. \tag{22}$$

Conditioned on a *known visible hit* $h_r = i$ at fixed pixel $r = (k, c, u)$, one valid additive-Gaussian observation equation is

$$\boxed{y_r = \begin{bmatrix} a_i \\ d_{i,c,k} \end{bmatrix} + v_r, \qquad v_r \sim \mathcal{N}(0, R_{r,i}), \qquad y_r = \begin{bmatrix} I_{k,c}(u) \\ D_{k,c}(u) \end{bmatrix}.} \tag{23}$$

The mean-depth pose Jacobian for the left error (18) is $H^{\mathrm{depth}}_{r,i} = e_3^\top R_{cw,c}[I_3 \; -[x_{i,k}]_\times]$; instantaneous twist columns are zero. For the *projected-point image-sampling* residual (4), let $J_\pi = \partial\pi/\partial m$ and $J_x = [I_3 \; -[x]_\times]$. Its distinct Jacobian is

$$H^{\mathrm{sample}}_{i,c,k} = \begin{bmatrix} \nabla I_{k,c}(u_i) J_\pi R_{cw,c} J_x & 0_{3\times 6} \\ (\nabla D_{k,c}(u_i) J_\pi - e_3^\top) R_{cw,c} J_x & 0_{1\times 6} \end{bmatrix}. \tag{24}$$

However, the *full* image output is not globally additive Gaussian after visibility/association is unknown. The actual nonlinear output law for all pixels is

$$\boxed{Y_k \mid (s_k, m, e, b) \sim \prod_{c,u} L_{(k,c,u)}(\cdot \mid s_k, m, e, b), \qquad L_r(y) = v_{r,\emptyset} f_{r,\emptyset}(y) + \sum_i v_{r,i} f_{r,i}(y).} \tag{25}$$

Here $v_{r,i}$ are the *normalized first-hit probabilities* (9), including geometry-dependent coverage and occlusion, and each f is a properly normalized distribution on the union of valid RGB-D returns and the separate no-return outcome. Equation (23) is only a conditional branch of (25). Neither (4) nor (23) alone may replace (25) in a convergence claim for full images.

6.3 Full measurement score and iterated Kalman correction

Let $\gamma_{r,j} = v_{r,j} f_{r,j}(y_r)/L_r(y_r)$ for j equal to any candidate or empty. In a smooth fixed depth-order region, with ϑ any continuous tangent parameter,

$$\boxed{\nabla_\vartheta \log L_r = \sum_j \gamma_{r,j} \left(\nabla_\vartheta \log v_{r,j} + \nabla_\vartheta \log f_{r,j}(y_r)\right).} \tag{26}$$

For $\eta_{r,j} = e_j c_{r,j}$, $\nabla \log v_{r,i} = \nabla\eta_{r,i}/\eta_{r,i} - \sum_{j\prec i} \nabla\eta_{r,j}/(1-\eta_{r,j})$ and $\nabla \log v_{r,\emptyset} = -\sum_j \nabla\eta_{r,j}/(1-\eta_{r,j})$ away from probability boundaries; stable limiting/prefix differentiation handles exact zero/one. The sensor-density derivative follows from (23) when its Gaussian branch applies. Therefore the coverage/visibility gradient is *part of the measurement Jacobian information* and must not be omitted.

The scene discrete state for a marginalized output convention is $z = (e, b)$; h_r is already summed out in L_r. Alternatively one can retain h_r as a complete-data latent variable and optimize $\mathbb{E}_q[\log v_{r,h_r} + \log f_{r,h_r}]$. *Do not use the complete-data latent h_r and the marginalized $\log L_r$ simultaneously.* Hold the scene posterior $q(z)$ and map fixed in Stage A. With tangent δ about the *fixed predicted* state define

$$E_k(\delta) = \tfrac{1}{2}\delta^\top (P_k^-)^{-1}\delta - \mathbb{E}_{q(z)} \sum_{c,u} \log L_{(k,c,u)}(y_{k,c,u} \mid \hat{s}_k^- \boxplus \delta, m, z). \tag{27}$$

Then $g_k = (P_k^-)^{-1}\delta - \nabla_\delta \ell_k(\delta)$ and the iterated information update is

$$\boxed{\left[(P_k^-)^{-1} - \nabla_\delta^2 \ell_k(\delta_j) + \lambda_j D_j\right]\Delta_j = -g_k(\delta_j), \qquad \delta_{j+1} = \delta_j + \alpha_j \Delta_j.} \tag{28}$$

The damping term makes the metric positive definite; accept the update against the *actual nonlinear* objective (27) and rebuild visibility after the accepted change. An explicitly labeled positive-definite Fisher/Gauss–Newton metric may replace the Hessian *as a local search metric*, never the full score. The same prediction prior and original pixels must be reused at every IEKF relinearization. For a known-association additive Gaussian special case, this reduces to the familiar $[(P^-)^{-1} + H^\top R^{-1} H]\Delta = H^\top R^{-1}(y - h) - (P^-)^{-1}\delta$. For an occlusion mixture the returned covariance is a *local Gaussian approximation*, not the exact generally multimodal posterior.

Backward Rauch–Tung–Striebel smoothing uses $C_k = P_{k|k} F_k^\top (P_{k+1|k})^{-1}$ and

$$\hat{s}_{k|K} = \hat{s}_{k|k} \boxplus C_k(\hat{s}_{k+1|K} \boxminus \hat{s}_{k+1|k}), \tag{29}$$

$$P_{k|K} = P_{k|k} + C_k(P_{k+1|K} - P_{k+1|k})C_k^\top, \tag{30}$$

with all errors and covariances transported into consistent Lie tangent charts. Static map uncertainty cannot be eliminated by pretending it is zero: retain joint map–pose covariance via state augmentation or algebraically exact information-block elimination. These equations are *the complete state/process/output/filter/smoother contract*; the outer alternating geometry stage remains defined by Sections 8 and 10.

7 Geometry discovery is a deterministic posterior computation

7.1 Posterior responsibilities rather than thresholds

For fixed (s, m, e, b), the exact conditional posterior that hit i generated ray r is

$$\gamma_{r,i} = \mathbb{P}(h_r = i \mid y_r, s, m, e, b) = \frac{v_{r,i} f_{r,i}(y_r)}{L_r(y_r)}, \quad \gamma_{r,\emptyset} = \frac{v_{r,\emptyset} f_{r,\emptyset}(y_r)}{L_r(y_r)}. \tag{31}$$

These values are *deterministic functions* of the current state and sensor distributions. They are not thresholds. When e, b are also uncertain, the relevant γ are obtained by averaging conditional expressions over their posterior, or by an explicitly declared variational approximation. The complete posterior generally couples many rays through shared points; independent per-ray soft assignments need not equal the exact posterior.

7.2 Backward consistency of a newly exposed candidate

For a candidate observed at time s, invert the current rigid motion and project to every other frame as in (8). Let $D_{k,c}$ denote earlier measured depth and $\hat{d}_{i,k,c}$ the new candidate's predicted depth at its projected location. The ordering probability under a Gaussian approximation with independent depth errors is

$$\mathbb{P}(\hat{d}_{i,k,c} > D_{k,c} \mid \text{uncertain depth}) = \Phi\left(\frac{\bar{d}_i - \bar{D}}{\sqrt{\sigma_i^2 + \sigma_D^2}}\right). \tag{32}$$

This is a useful pairwise diagnostic, *not* a replacement for the full ray likelihood (12). A previously hidden point need not match the foreground RGB. A previously visible surface should either match earlier RGB-D or lower the model likelihood; an out-of-view ray gives no evidence.

7.3 Birth as an evidence ratio

For a candidate i, let $H_1 : e_i = 1$ and $H_0 : e_i = 0$. Given other geometry, poses, and the original sequence, the posterior odds are

$$\boxed{\frac{\mathbb{P}(e_i = 1 \mid \mathcal{Y}, \text{others})}{\mathbb{P}(e_i = 0 \mid \mathcal{Y}, \text{others})} = \frac{\mathbb{P}(e_i = 1 \mid \text{others})}{\mathbb{P}(e_i = 0 \mid \text{others})} \frac{p(\mathcal{Y} \mid e_i = 1, \text{others})}{p(\mathcal{Y} \mid e_i = 0, \text{others})}.} \tag{33}$$

Each likelihood covers *all original rays once* and integrates unknown p_i, a_i, b_i as required. If the candidate is always hidden before its first observation, those earlier rays provide little evidence against existence. If it would repeatedly sit in visible free space, the Bayes factor can strongly disfavor it. The prior odds ensure that giving each observed pixel its own new surface is not automatically rewarded. All quantities are statistical parameters or inferred probabilities, not ad hoc cutoffs.

A candidate observed only once may have little evidence beyond its source ray and map prior. To prevent the birth frame from "validating itself twice," the source ray is used once in the joint likelihood; a backprojected initialization is *not an additional observation factor*. A source-excluded predictive Bayes factor can be used diagnostically, but must not be added as a second training loss on top of the original ray likelihood.

7.4 Canonical map refinement and uncertainty

Under a currently hypothesized part b, a measured point at time s yields initial canonical coordinate $p_i = T_{b,s}^{-1} y_{i,s}^{3D}$. If $\delta p_i = J_y \delta y + J_T \delta \xi_{b,s}$, then (assuming initial independence)

$$P_{p_i} = J_y R_y J_y^\top + J_T P_T J_T^\top, \qquad P_{p_i,T} = J_T P_T. \tag{34}$$

The cross-covariance is essential. It prevents several points initialized from one uncertain motion from being treated as independent confirmations of that same motion. A batch smoother naturally preserves the coupling through shared pose and point variables; an efficient implementation may eliminate map increments through a Schur complement.

8 One common posterior objective

Let $\theta = (s, m)$ denote the continuous variables, and z all discrete variables. Choose proper priors $p(s), p(m), p(z)$ (existence/part/complexity included) and the valid-ray and invalid-ray likelihoods above. Define

$$p(\mathcal{Y}, z, \theta) = p(\theta, z) \prod_{r \in \mathcal{R}} p(y_r \mid z, \theta), \quad \ell(\theta) = \log \sum_z p(\mathcal{Y}, z, \theta). \tag{35}$$

The product is a *conditional* sensor model: shared uncertainties from camera calibration or correlated depth errors must be explicitly represented if present. With a probability mass/density $q(z)$ in an admissible family $\mathcal{Q}$, the evidence lower bound (ELBO) is

$$\boxed{\mathcal{F}(q,\theta) = \mathbb{E}_q[\log p(\mathcal{Y}, z, \theta)] + H(q), \qquad H(q) = -\mathbb{E}_q \log q(z).} \tag{36}$$

For the unrestricted family $\mathcal{Q}$ of all distributions on finite z,

$$\boxed{\ell(\theta) = \mathcal{F}(q,\theta) + \mathrm{KL}(q \| p(z \mid \mathcal{Y}, \theta)).} \tag{37}$$

Thus $\mathcal{F} \le \ell$ with equality after an exact posterior update. This identity makes the two-stage decomposition precise:

$$\textbf{A} - \textbf{motion:} \quad s^{r+1} \text{ increases } \mathcal{F}(q^r, s, m^r), \tag{38}$$

$$\textbf{B} - \textbf{geometry:} \quad (m^{r+1}, q^{r+1}) \text{ increase } \mathcal{F}(q, s^{r+1}, m). \tag{39}$$

The geometry stage includes both updating point parameters and deterministic inference of posterior existence, part, and correspondence probabilities. If one uses a tractable restricted $\mathcal{Q}$ (e.g. mean-field), the proof concerns the *declared ELBO*, not necessarily the exact marginal log posterior ℓ. In either case motion is frozen during B, and geometry is frozen during A.

9 A complete algorithm without geometry thresholds

Algorithm 1 Occlusion-consistent bidirectional RGB-D IEKS (finite catalogue)

Require: Original calibrated RGB-D images $\mathcal{Y}$; finite candidate catalogue from observed depth rays; calibrated measurement and coverage laws; proper priors; fixed pose gauge

Initialize candidate canonical geometry by inverse transporting source observations; initialize poses and probabilities, without declaring candidates accepted/rejected
Construct one full probabilistic ray model $p(\mathcal{Y}, z, \theta)$
for $r = 0, 1, 2, \ldots$ **do**
    Stage A (rigid motion, m, q fixed):
    Build analytic $SE(3)$ Jacobians and full motion objective gradient
    Compute iterated forward IEKF / backward RTS smoothing direction, or an equivalent sparse information solve
    Accept a damped Lie-retraction step only if the common ELBO does not decrease
    Stage B (geometry, s fixed):
    Reproject candidate points from their acquisition times into earlier and later RGB-D frames
    Optimize canonical points, appearance and optional surfel support with current probabilities fixed; accept only a nondecreasing ELBO step
    Update posterior/variational probabilities of point existence, rigid-part ownership, and ray associations using all RGB-D measurements
    Refresh occlusion ordering, coverage, and all image-space visibility terms
end for
return One shared canonical map with existence/part probabilities; rigid poses/twists at all frames; motion and geometric uncertainty

The loop is stopped in practice using a numerical optimization-tolerance criterion. This is a solver termination tolerance, not a decision threshold distinguishing new geometry from moving geometry. The theory below concerns asymptotic iterates.

10 Convergence: statements and rigorous proofs

10.1 Assumptions: what exactly is being proved

Assumption 10.1 (Finite, proper model)**.** The candidate and component hypothesis sets are finite; the ray sensor densities, priors, and any missing-depth probabilities are normalized and strictly positive on the data under consideration. Measurement covariance eigenvalues are bounded above and below by positive constants. The gauge is fixed, and the relevant upper level set of $\mathcal{F}$ is compact. The admissible variational family $\mathcal{Q}$ is compact (full finite simplex, or a compact closed factorized family).

Assumption 10.2 (Exact merit comparison)**.** Each A-step returns s^{r+1} satisfying $\mathcal{F}(q^r, s^{r+1}, m^r) \geq \mathcal{F}(q^r, s^r, m^r)$. Each B-step returns (q^{r+1}, m^{r+1}) satisfying $\mathcal{F}(q^{r+1}, s^{r+1}, m^{r+1}) \geq \mathcal{F}(q^r, s^{r+1}, m^r)$. All these comparisons evaluate the *same* fixed-data objective with its current visibility/coverage, not a different surrogate with changed normalization or repeated data.

Assumption 10.3 (Regularity for stationarity only)**.** Near each accumulation point, the active camera projection domains, depth ordering, and finite supports are locally fixed, or the forward visibility model is continuously smoothed. In a local chart the ELBO is continuously differentiable with block-Lipschitz gradients on the level set. The interior continuous coordinate blocks are visited every cycle; a boundary solution is interpreted via the usual constrained first-order/KKT condition.

Assumption 10.4 (Sufficiently improving updates for stationarity only)**.** For each continuous block $x \in \{s, m\}$, its actual update is a gradient-related ascent step of $\mathcal{F}$ with an Armijo backtracking rule, or another update ensuring $\mathcal{F}(\text{after}) - \mathcal{F}(\text{before}) \geq c_x \|\nabla_x \mathcal{F}(\text{before})\|^2$ for a fixed $c_x > 0$. For example $d_x = B_x^{-1} \nabla_x \mathcal{F}$ with eigenvalues of B_x bounded in $[\mu_x, M_x]$, $0 < \mu_x \leq M_x < \infty$, plus a standard Armijo rule under the uniform Lipschitz bound. The variational block is exactly maximized over $\mathcal{Q}$ at each cycle, or cyclically optimized by a closed, strictly improving rule whose limit points obey its block KKT conditions. Consecutive accepted increments tend to zero (this follows for the gradient-related continuous steps from the sufficient-improvement bound; the property for an approximate q update must additionally hold).

Theorem 10.5 (Monotonic improvement and convergence of objective values)**.** *Under Assumptions 10.1–10.2, the complete sequence of ELBO values $\mathcal{F}(q^r, s^r, m^r)$ is nondecreasing and converges to a finite value $\mathcal{F}_\infty$. The parameter iterates have at least one accumulation point. No assertion of uniqueness or convergence to a global optimum follows.*

Proof. Applying the A-step and then the B-step yields

$$\begin{aligned} \mathcal{F}(q^r, s^r, m^r) &\leq \mathcal{F}(q^r, s^{r+1}, m^r) \\ &\leq \mathcal{F}(q^{r+1}, s^{r+1}, m^{r+1}). \end{aligned}$$

Thus the scalar ELBO sequence is monotone. By Assumption 10.1, the upper level set is compact and $\mathcal{F}$ is bounded above on it (bounded measurement densities/prior on a compact parameter set and finite latent entropy). The monotone convergence theorem for real sequences implies $\mathcal{F}_r \uparrow \mathcal{F}_\infty < \infty$. Compactness gives accumulation points of parameter iterates. Monotonicity says nothing about the identity of such points and does not imply that the iterates themselves converge. □

Corollary 10.6 (Exact E-step also improves the marginal objective)**.** *If $\mathcal{Q}$ is unrestricted and each geometry stage ends with $q^{r+1} = p(z \mid \mathcal{Y}, \theta^{r+1})$ (after its continuous updates), and q^r is exact at the start of motion stage, then the marginal log posterior $\ell(\theta^r)$ is nondecreasing at completed cycles.*

Proof. The identity (37) and exact posterior updates give

$$\ell(\theta^r) = \mathcal{F}(q^r, \theta^r) \le \mathcal{F}(q^{r+1}, \theta^{r+1}) = \ell(\theta^{r+1}).$$

The inequality is Theorem 10.5. This is generalized-EM monotonicity, not a claim that the IEKS solution is an exact optimizer. □

Lemma 10.7 (Uniform ascent from a damped information direction)**.** *Suppose in a regular chart the block gradient $g = \nabla_x \mathcal{F}$ is L-Lipschitz, $B = B^\top \succ 0$ has eigenvalues in $[\mu, M]$, and $d = B^{-1}g$. Then $g^\top d \ge \|g\|^2/M$, $\|d\| \le \|g\|/\mu$, and an Armijo backtracking line search with fixed contraction and slope parameter $\eta \in (0,1)$ accepts a step length bounded away from zero as long as these constants hold uniformly. Consequently the accepted increase is bounded below by a positive multiple of $\|g\|^2$.*

Proof. The eigenvalue inequalities give the two displayed bounds. The ascent/descent lemma gives $\mathcal{F}(x + \alpha d) \ge \mathcal{F}(x) + \alpha g^\top d - \frac{L}{2}\alpha^2\|d\|^2$. Armijo ascent requires an increase of at least $\eta\alpha g^\top d$, and is therefore guaranteed whenever $\alpha \le 2(1-\eta)(g^\top d)/(L\|d\|^2) \ge 2(1-\eta)\mu^2/(LM)$. With a fixed geometric backtracking factor the first accepted step is no smaller than a fixed positive fraction of the latter positive bound (capped at the initial trial length). Combining $g^\top d \ge \|g\|^2/M$ with the accepted Armijo inequality yields the claim. Retraction coordinates give the manifold version locally when their differential and smoothness bounds are uniform. □

Theorem 10.8 (Stationarity of accumulation points: local, conditional)**.** *Under Assumptions 10.1–10.4, every accumulation point (q^*, s^*, m^*) of the* cycle-boundary *iterates satisfies*

$$\nabla_s \mathcal{F}(q^*, s^*, m^*) = 0, \qquad \nabla_m \mathcal{F}(q^*, s^*, m^*) = 0$$

in free continuous coordinates (or the corresponding constrained first-order conditions). If the variational block is exactly maximized, q^* maximizes $\mathcal{F}(q, s^*, m^*)$ over $q \in \mathcal{Q}$. For coordinate-wise variational updates, only the corresponding coordinate-wise KKT statement follows. This is stationarity of the chosen ELBO*, not global posterior optimality.*

Proof. Let $\Delta_r = \mathcal{F}_{r+1} - \mathcal{F}_r \ge 0$. Theorem 10.5 implies $\Delta_r \to 0$. Each substep increase is nonnegative and bounded by Δ_r, hence also tends to zero. The sufficient-improvement condition for the motion update implies $\|\nabla_s \mathcal{F}(q^r, s^r, m^r)\|^2 \le \Delta_r/c_s \to 0$. For the map update (performed at fixed q^r before the posterior refresh), it similarly implies $\|\nabla_m \mathcal{F}(q^r, s^{r+1}, m^r)\| \to 0$ at its pre-update values. Because the continuous accepted increments tend to zero and the variational update increments tend to zero by Assumption 10.4, the intermediate block states and cycle-boundary states share each accumulation point. Continuity of both gradients then establishes their vanishing at that point (with projected-gradient conditions at constrained boundaries). For an exactly maximized variational update, $\mathcal{F}(q^{r+1}, s^{r+1}, m^{r+1}) \ge \mathcal{F}(q, s^{r+1}, m^{r+1})$ for all $q \in \mathcal{Q}$. Pass to a convergent subsequence and use continuity to conclude the maximizing property for q^*. Approximate variational coordinate updates yield only their own blockwise necessary conditions by the same closed-update argument. The argument does not establish a unique limit point or exclude local saddles. □

Remark 10.9 (What must be checked in an implementation)**.** An IEKS step built from an incomplete Jacobian may not be an ascent direction of the *full* ray objective. A positive-definite information approximation multiplied by the *full actual gradient*, together with an actual-objective line search, is sufficient. If a proposed direction fails to increase the objective, replace it by a damped full-gradient direction or reject it. The stationarity theorem needs uniform gradient-related progress, not merely the absence of observed decreases. Hard changes in depth ordering may violate differentiability exactly at a limiting tie; only monotone *value* convergence remains unconditional across such changes.

11 Identifiability, irreducible ambiguity, and limits of convergence

Proposition 11.1 (A newly exposed point observed once cannot determine its owner from geometry alone)**.** *Suppose candidate point i is observed at only one time s, with exact 3D location y_s and unknown canonical location p_i. For* each *candidate rigid-component trajectory $T_{b,k}$, the choice $p_i = T_{b,s}^{-1} y_s$ reproduces its source observation exactly. If all other frames are either out of view or hide the candidate under each proposed trajectory, RGB-D observations alone cannot identify b or its unseen positions.*

Proof. For every b, substitute $p_i = T_{b,s}^{-1} y_s$ into $T_{b,s} p_i = y_s$. At other frames, the stated lack of visibility makes the candidate absent from visible-ray likelihood contributions; hence these frames cannot distinguish among the hypotheses. Any resulting posterior preference arises from priors or additional constraints, not data identifiability. □

For two matched observations at k and l, inverse-transported coordinates must agree for the correct body:

$$T_{b,k}^{-1} y_{i,k}^{3D} - T_{b,l}^{-1} y_{i,l}^{3D} \approx 0, \tag{40}$$

with covariance inherited from depth and motion. Thus repeated appearances, nonconsecutive matches, RGB texture, varied surface normals, and free-space constraints can supply information that the initial view lacks. A large collection of identical planar or symmetric surfaces still admits ambiguous solutions.

Proposition 11.2 (Why motion and map cannot be freely optimized together)**.** *If each measurement $y_{k,j}^{3D}$ is allowed its own unconstrained canonical point $p_{k,j}$ with no map-complexity prior and is required to match only its source frame, then for* any *rigid trajectory T_k the selection $p_{k,j} = T_k^{-1} y_{k,j}^{3D}$ achieves zero geometric source-frame residual. Hence reconstruction alone supplies no motion identification.*

Proof. Immediate from $T_k p_{k,j} = y_{k,j}^{3D}$ for every observation. Only cross-frame prediction, shared finite geometry, occlusion reasoning, and priors disambiguate motion. □

Convergence is not completeness. Theorem 10.8 concerns convergence to a stationary solution of a particular finite probabilistic model under restrictive optimization assumptions. It does *not* prove that all points become correctly assigned, that the number of components is recovered, that an unobserved surface can be reconstructed, or that the true physical motion is the unique global optimum. Recovering true geometry would additionally need statistical identifiability, appropriate sensor model, sufficiently informative views, and global inference; those are not guaranteed by a local IEKS convergence theorem.

12 Computational consequences and validation plan

12.1 Why the state remains computationally manageable

The motion block contains $6(B+1)(K+1)$ pose coordinates (plus twists), whereas the candidate map can be large. With fixed correspondences and active rays, the joint linearized normal equations have blocks

$$\begin{bmatrix} A_{ss} & A_{sm} \\ A_{ms} & A_{mm} \end{bmatrix} \begin{bmatrix} \delta s \\ \delta m \end{bmatrix} = \begin{bmatrix} b_s \\ b_m \end{bmatrix}, \qquad (A_{ss} - A_{sm} A_{mm}^{-1} A_{ms}) \delta s = b_s - A_{sm} A_{mm}^{-1} b_m. \tag{41}$$

This Schur complement eliminates large map increments *computationally* while preserving their information about shared motion. Near-independent surfels yield approximately block-sparse A_{mm}. The information matrix need not be formed densely: matrix-free $v \mapsto H^\top R^{-1}(Hv)$ products are possible, as in the original sparse IEKF note.

12.2 Theory-grounded ablations

The implementation should separately record (i) the *same* total ray ELBO/marginal posterior across cycles, (ii) its motion-block and geometry-block increases, (iii) forward/backward image and depth prediction likelihoods, (iv) calibrated posterior existence and part probabilities, (v) known-mask occlusion/free-space contradictions, and (vi) rank/conditioning of the rigid-state information. Test an initially occluded moving drawer, a newly revealed *static* interior, a textureless symmetric part, a point visible only once, nonconsecutive views, and deliberate incorrect initial poses. A monotonic negative-log objective is necessary for the stated optimization theorem but not sufficient for ground-truth correctness.

13 Conclusion

The proposed method is *one* bidirectional, occlusion-aware RGB-D point model. Stage A estimates rigid transport analytically via an information-form iterated extended Kalman smoother, holding current geometric hypotheses fixed. Stage B freezes this motion, reprojects later observations backward, and deterministically computes their probabilistic support as existing, newly represented, hidden, or sensor-corrupted surface hypotheses. Previously revealed observations re-enter the next motion pass through the *same* rigid transforms; thus information from late geometry can revise earlier motion. The normalized ray likelihood is the missing theoretical ingredient connecting direct projection, RGB/depth residuals, occlusion, and geometric discovery without hand-selected residual thresholds. Under the explicitly stated assumptions, alternating ELBO improvements converge in value and suitably regular, sufficiently improving iterates have stationary accumulation points. Nothing stronger is proved without additional observability and global-inference assumptions.

References

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